

FULL ENTROPY DIMENSION FOR A COUNTABLE STRONGLY OVERLAPPING 3-ADIC RENEWAL MEASURE

ERIC GROVER

ABSTRACT. Let $K_0, K_1, \dots$ be independent with $\Pr(K = k) = 2^{-k}, k \geq 1$, put $S_j = K_0 + \dots + K_j$, and let ν be the law of

$$X = \sum_{j \geq 0} 3^j 2^{-S_j} \in \mathbb{Z}_3.$$

Equivalently ν is the stationary measure of the countably infinite iterated function system $\phi_k(x) = (1 + 3x)/2^k$ with weights 2^{-k} , whose images overlap heavily. We prove that the level- n collision number $R_n = 3^n \sum_c \Pr(X \equiv c)^2$ satisfies $R_n \leq 82n^3$, hence that the entropy deficit is $O(\log n)$ and ν has full entropy dimension in $\mathbb{Z}_3$. The proof is elementary: at a fixed renewal total, distinct words in one 3-adic cylinder are separated on the real line, and decomposing simultaneously by renewal total and by dyadic real scale converts that separation into an L^2 estimate whose two aggregate charges are bounded by 1 and $n/6$.

We then show the method is rigid. Within the natural family indexed by coprime $p, q \geq 2$ and a geometric gap parameter λ , the three hypotheses the proof requires hold simultaneously for exactly one triple, namely $(p, q, \lambda) = (3, 2, 1/2)$, and all three hold with equality. We stress that the method delivers a polynomial of degree three whereas machine-precision computation of R_n to level 14, from the exact recurrence, indicates the truth is degree one. We isolate the two independent factors of n responsible, record the two statements that would close the gap, and identify the structural reason neither is reachable here: the stationary real address has critical moment exponent exactly 1, so it possesses no moment of order ≥ 1 , and every geometric weighting of the scale decomposition is therefore unavailable.

1. INTRODUCTION

1.1. The measure. All logarithms are natural. Let K have the geometric law

$$\Pr(K = k) = 2^{-k}, \quad k = 1, 2, \dots, \quad (1)$$

let $K_0, K_1, \dots$ be independent copies of K , and set $S_j = K_0 + \dots + K_j$. The j -th term of

$$X = \sum_{j \geq 0} 3^j 2^{-S_j} \quad (2)$$

has 3-adic valuation j , so the series converges in $\mathbb{Z}_3$. Write ν for its law. Since $X \equiv 2^{-K_0} \not\equiv 0 \pmod{3}$, the measure ν is supported on $\mathbb{Z}_3^\times$.

Equivalently, ν is the unique stationary measure of the iterated function system

$$\phi_k(x) = \frac{1 + 3x}{2^k}, \quad \Pr(\phi_k) = 2^{-k}, \quad k \geq 1. \quad (3)$$

Every map contracts by exactly $1/3$ in the 3-adic metric, so the similarity dimension is $h/\log 3$ with $h = H(K) = 2 \log 2$, giving $2 \log 2 / \log 3 \approx 1.26 > 1$. The system is therefore supercritical, and the question is whether the overlaps destroy entropy. Two features place it outside the

2020 *Mathematics Subject Classification.* Primary 28A80; Secondary 11K41, 37A44, 94A17.

Key words and phrases. self-similar measure, iterated function system with overlaps, p -adic dynamics, entropy dimension, renewal process, collision entropy.

existing theory: the alphabet is countably infinite, and the overlaps are severe, since ϕ_k and $\phi_{k'}$ have images meeting in sets of full relative measure.

For $n \geq 1$ only the first n terms of (2) affect the residue modulo 3^n . Put

$$C_n = \sum_{j=0}^{n-1} 3^j 2^{-s_j} \bmod 3^n, \quad p_n(c) = \Pr(C_n = c), \quad (4)$$

and define the collision number and the entropy deficit

$$R_n = 3^n \sum_{c \bmod 3^n} p_n(c)^2, \quad D_n = n \log 3 - H(C_n). \quad (5)$$

1.2. Results.

Theorem 1.1. *For every $n \geq 1$,*

$$R_n \leq 70n^2 \left(1 + \frac{n}{6}\right) + 2\gamma^n, \quad \gamma = \frac{3^{13}}{5^{10}} < 1. \quad (6)$$

In particular $R_n \leq 82n^3$ and

$$D_n \leq \log 82 + 3 \log n = o(n), \quad (7)$$

so that $H(C_n)/(n \log 3) \rightarrow 1$: the measure ν has full entropy dimension in $\mathbb{Z}_3$.

Sections 2–5 prove Theorem 1.1. Section 6 proves that the method reaches exactly one system, Section 7 quantifies how far $O(n^3)$ is from the truth, and Section 8 records an equivalent affine-correlation form.

1.3. Relation to the literature. The entropy and dimension theory of overlapping self-similar measures on $\mathbb{R}$ is by now substantial, but it does not cover the present system, for two independent reasons.

First, *finiteness*. Hochman’s inverse entropy theorem [5] and its descendants—Shmerkin’s L^q formula under exponential separation [11], Varjú’s results for transcendental and algebraic parameters [12, 13], Breuillard–Varjú [2, 3], Rapaport [9]—are all stated for a *finite* iterated function system; the recent survey [14] defines an IFS as a finite collection in its opening sentence. Countable conformal systems do have a dimension theory, due to Mauldin and Urbański [6], but it assumes the open set condition. With overlaps and a countable alphabet one has pressure and overlap-function estimates [7, 8], not an identification with the similarity dimension.

Second, *effectivity*. Even where the real-line theory applies, it is not quantitative. Hochman remarks explicitly, immediately after his inverse theorem, that although the dependence of δ on (ε, m) is effective, “the value of n depends among other things on the rate at which $H_m(\mu) \rightarrow \dim \mu$, which is currently not effective” [5, §2]. Consequently no bound of the shape “entropy deficit at scale n is at most $f(n)$ ” with explicit f follows from that theory. Theorem 1.1 is of exactly that shape, with every constant written down.

In the p -adic setting, Hare and Vávra [4] develop self-similar sets and measures in $\mathbb{Q}_p$, but for a *finite* system whose contraction ratios are integer powers of p and whose translations are rational; their theory is of finite type and automata-theoretic, and their open questions do not concern overlaps in the present sense. Earlier p -adic constructions [1] are graph-directed and separated. To the author’s knowledge no ultrametric analogue of the inverse entropy theorem exists. The entropy technology has been transported to $\mathbb{Q}_p$ for the discretized sum-product problem [10], but not to self-similar measures.

The measure (2) arose in an unrelated investigation and is studied here on its own terms; nothing below bears on that origin.

1.4. Acknowledgements. The numerical work reported in Section 7 and the verification scripts described in Appendix A were carried out with computational assistance. All statements labelled as theorems, lemmas or propositions are proved in full in the text and do not depend on any computation.

2. FINITE RENEWAL WORDS AND REAL SEPARATION

Let $w = (k_0, \dots, k_{n-1}) \in \mathbb{Z}_{\geq 1}^n$ be a renewal word. Put

$$s_j(w) = k_0 + \dots + k_j, \quad t(w) = s_{n-1}(w), \quad (8)$$

and define the *finite address*

$$x_n(w) = \sum_{j=0}^{n-1} 3^j 2^{-s_j(w)} = \frac{N_n(w)}{2^{t(w)}}, \quad N_n(w) = \sum_{j=0}^{n-1} 3^j 2^{t(w)-s_j(w)} \in \mathbb{Z}. \quad (9)$$

Note that $x_n(w)$ is simultaneously a positive rational number and, since 2 is a unit in $\mathbb{Z}_3$, an element of $\mathbb{Z}_3$ reducing to the cylinder address of w . This double reading is the engine of the whole argument. The probability of w is

$$\Pr(w) = \prod_{j=0}^{n-1} 2^{-k_j} = 2^{-t(w)}, \quad (10)$$

which depends on w only through its total.

Lemma 2.1 (Word recovery). *The pair $(N_n(w), t(w))$ determines w uniquely. In particular distinct words of the same total have distinct addresses.*

Proof. For $n = 1$ we have $N_1(w) = 1$ identically and $t(w) = k_0$, so t alone determines w . Let $n \geq 2$. In (9) the summand $j = n - 1$ equals 3^{n-1} , which is odd, while every summand with $j < n - 1$ has exponent $t - s_j \geq k_{n-1} \geq 1$ and is even; hence $N_n(w)$ is odd. Moreover

$$N_n(w) - 3^{n-1} = \sum_{j=0}^{n-2} 3^j 2^{t-s_j},$$

and every exponent here is at least k_{n-1} . Factoring out $2^{k_{n-1}}$ and using $t - k_{n-1} = s_{n-2}$ leaves $\sum_{j \leq n-2} 3^j 2^{s_{n-2}-s_j}$, whose $j = n - 2$ term is the odd number 3^{n-2} and whose earlier terms are even. Therefore

$$k_{n-1} = v_2(N_n(w) - 3^{n-1}), \quad (11)$$

and the quotient $(N_n(w) - 3^{n-1})/2^{k_{n-1}}$ is precisely the numerator attached to the prefix $(k_0, \dots, k_{n-2})$, whose total is $t - k_{n-1}$. Induction recovers every increment. $\square$

Lemma 2.2 (Fixed-total cylinder separation). *Let $u \neq v$ be words of length n with the same total t . If $x_n(u) \equiv x_n(v) \pmod{3^n}$ then*

$$|x_n(u) - x_n(v)| \geq \frac{2 \cdot 3^n}{2^t} =: \Delta_{n,t}. \quad (12)$$

Proof. The common denominator 2^t is a unit modulo 3^n , so the hypothesis gives $3^n \mid N_n(u) - N_n(v)$. By Lemma 2.1 that difference is nonzero, and by the proof of Lemma 2.1 both numerators are odd, so the difference is even. As $\gcd(2, 3^n) = 1$ we get $|N_n(u) - N_n(v)| \geq 2 \cdot 3^n$, and division by 2^t gives (12). $\square$

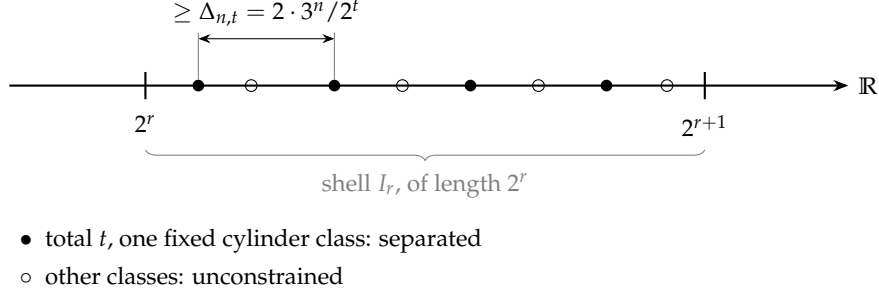


FIGURE 1. Lemma 2.2. Within one 3-adic cylinder and one renewal total, real addresses are $\Delta_{n,t}$ -separated, so a dyadic shell of length 2^r carries at most $1 + 2^r / \Delta_{n,t}$ of them. Addresses in different classes are subject to no such constraint.

Finally, since $s_j(w) \geq j + 1$,

$$0 < x_n(w) \leq \sum_{j=0}^{n-1} \frac{3^j}{2^{j+1}} = \left(\frac{3}{2}\right)^n - 1. \quad (13)$$

3. DECOMPOSITION BY TOTAL AND BY REAL SCALE

Fix n . For an integer $t \geq n$ and $r \in \mathbb{Z}$ let $I_r = [2^r, 2^{r+1})$ and define the subprobability on $\mathbb{Z}/3^n\mathbb{Z}$

$$p_{t,r}(c) = \sum_{\substack{w \in \mathbb{Z}_{\geq 1}^n: t(w)=t \\ x_n(w) \in I_r, x_n(w) \equiv c \pmod{3^n}}} 2^{-t}, \quad \varepsilon_{t,r} = \|p_{t,r}\|_1. \quad (14)$$

We refer to the pairs (t, r) as *cells*.

Lemma 3.1 (Cell atom bound). *For every cell,*

$$\|p_{t,r}\|_\infty \leq 2^{-t} + \frac{2^r}{2 \cdot 3^n}, \quad \text{hence} \quad 3^n \|p_{t,r}\|_2^2 \leq (3^n 2^{-t} + 2^{r-1}) \varepsilon_{t,r}. \quad (15)$$

Proof. By Lemma 2.2 the addresses of a fixed total lying in one residue class are mutually separated by at least $\Delta_{n,t}$, so the interval I_r , of length 2^r , contains at most $1 + 2^r / \Delta_{n,t}$ of them. Each carries weight 2^{-t} by (10), which is the first claim. The second follows from $\sum_c p_{t,r}(c)^2 \leq \|p_{t,r}\|_\infty \|p_{t,r}\|_1$. $\square$

We retain all totals in the range

$$n \leq t < 6n, \quad (16)$$

which we call the *core*; there is no lower-tail deletion. If $t < 6n$ then $x_n(w) \geq 2^{-k_0} \geq 2^{-t} > 2^{-6n}$, since $k_0 \leq t$. Together with (13) this confines core addresses to shells with

$$-6n \leq r < n. \quad (17)$$

There are at most $7n$ such shells and exactly $5n$ totals in (16), so the number Q_n of nonempty core cells obeys

$$Q_n \leq 35n^2. \quad (18)$$

Remark 3.2. Inequality (17) is slightly generous: (13) gives $r \leq \lfloor n \log_2(3/2) \rfloor$, so the exact count of admissible cells is

$$Q_n \leq 5n(6n + \lfloor n \log_2(3/2) \rfloor + 1) \leq 32.925n^2 + 5n.$$

This is asymptotically $32.93n^2$, but it is *not* bounded by $33n^2$ for small n : at $n = 1, 2, 3, 4, 6, \dots$ the left side exceeds $33n^2$, and it equals $35n^2$ exactly at $n = 1$ and $n = 2$. The clean bound (18) is therefore the one to use. What equals $35n^2$ at $n = 1, 2$ is the rectangular range count above, not the number Q_n of nonempty cells, which is far smaller ($Q_1 = 5$, $Q_2 = 46$; see Section 7); replacing it by the asymptotic form would gain nothing in Theorem 1.1, whose constant is determined entirely by $n = 1$.

4. THE TWO AGGREGATE CHARGES

The point of the decomposition is that the two terms of (15) aggregate over all cells with no loss exponential in n .

Lemma 4.1 (Diagonal charge). $3^n \sum_{t,r} 2^{-t} \varepsilon_{t,r} \leq 1$, the sum being over core cells.

Proof. The total $T_n = S_{n-1}$ is negative binomial, $\Pr(T_n = t) = \binom{t-1}{n-1} 2^{-t}$ for $t \geq n$, and the shells partition the words of a fixed total, so $\sum_r \varepsilon_{t,r} \leq \Pr(T_n = t)$. Hence

$$3^n \sum_{t,r} 2^{-t} \varepsilon_{t,r} \leq 3^n \sum_{t \geq n} \binom{t-1}{n-1} 4^{-t} = 3^n \left(\frac{1/4}{1-1/4} \right)^n = 3^n 3^{-n} = 1,$$

using $\sum_{t \geq n} \binom{t-1}{n-1} z^t = (z/(1-z))^n$ at $z = 1/4$. □

Lemma 4.2 (Shell charge). $\sum_{t,r} 2^{r-1} \varepsilon_{t,r} \leq \frac{n}{6}$.

Proof. On I_r one has $2^r \leq x_n(w)$, so $\sum_{t,r} 2^{r-1} \varepsilon_{t,r} \leq \frac{1}{2} \mathbb{E}[x_n(W)]$. Now $\mathbb{E}[2^{-K}] = \sum_{k \geq 1} 4^{-k} = 1/3$, so by independence $\mathbb{E}[2^{-S_j}] = 3^{-(j+1)}$ and

$$\mathbb{E}[x_n(W)] = \sum_{j=0}^{n-1} 3^j \cdot 3^{-(j+1)} = \frac{n}{3}. \quad (19)$$

□

Remark 4.3 (The crux). Identity (19) is the whole theorem, and it is worth isolating why. The address has *exponential range* but *linear mean*: x_n may be as large as $(3/2)^n - 1$ by (13), yet $\mathbb{E}[x_n] = n/3$ because each term contributes exactly $3^j \cdot 3^{-(j+1)} = 1/3$. High shells therefore carry exponentially little mass, which is precisely what defeats the exponentially large factor 2^{r-1} in (15). An argument using only the sup bound (13) in place of the first moment yields nothing: it produces the factor $2^{r-1} \leq 2^{0.585n}$ in every cell. This exact balance is also what fails for every other parameter choice, as Section 6 shows.

Proposition 4.4 (Core collision bound). Let p_n^{core} be the subprobability contributed by the totals $n \leq T_n < 6n$ and $R_n^{\text{core}} = 3^n \|p_n^{\text{core}}\|_2^2$. Then

$$R_n^{\text{core}} \leq 35n^2 \left(1 + \frac{n}{6} \right). \quad (20)$$

Proof. By Lemma 3.1 and Lemmas 4.1, 4.2,

$$\sum_{t,r} 3^n \|p_{t,r}\|_2^2 \leq 1 + \frac{n}{6}. \quad (21)$$

Writing $a_{t,r} = 3^{n/2} \|p_{t,r}\|_2$, Minkowski's inequality and then Cauchy–Schwarz over the at most Q_n nonempty cells give

$$\sqrt{R_n^{\text{core}}} = 3^{n/2} \left\| \sum_{t,r} p_{t,r} \right\|_2 \leq \sum_{t,r} a_{t,r} \leq \left(Q_n \sum_{t,r} a_{t,r}^2 \right)^{1/2}.$$

Now apply (18) and (21) and square. $\square$

5. UPPER TAIL AND PROOF OF THEOREM 1.1

Since $\mathbb{E}[z^K] = z/(2-z)$ for $0 < z < 2$, exponential Markov at $z = 5/3$ gives

$$\Pr(T_n \geq 6n) \leq z^{-6n} \left(\frac{z}{2-z} \right)^n = \left[\left(\frac{3}{5} \right)^6 \cdot 5 \right]^n = \left(\frac{729}{3125} \right)^n. \quad (22)$$

Writing p_n^{tail} for the subprobability from $T_n \geq 6n$ and using $\|p\|_2^2 \leq \|p\|_1^2$,

$$R_n^{\text{tail}} \leq 3^n \Pr(T_n \geq 6n)^2 \leq \left[3 \left(\frac{729}{3125} \right)^2 \right]^n = \gamma^n, \quad \gamma = \frac{1594323}{9765625} = \frac{3^{13}}{5^{10}}. \quad (23)$$

Numerically $\gamma \approx 0.16325868 < 1$. (Any $z \in (1, 2)$ works in (22); $z = 5/3$ is chosen for arithmetic convenience.)

Since $p_n = p_n^{\text{core}} + p_n^{\text{tail}}$, the L^2 triangle inequality and $(a+b)^2 \leq 2a^2 + 2b^2$ give

$$R_n \leq 2R_n^{\text{core}} + 2R_n^{\text{tail}} \leq 70n^2 \left(1 + \frac{n}{6} \right) + 2\gamma^n,$$

which is (6). For the cubic form, divide by n^3 :

$$\frac{70n^2(1 + n/6) + 2\gamma^n}{n^3} = \frac{35}{3} + \frac{70}{n} + \frac{2\gamma^n}{n^3}.$$

Each of the last two terms is strictly decreasing in $n \geq 1$, so the right side attains its maximum at $n = 1$, where it equals

$$\frac{245}{3} + 2\gamma < \frac{245}{3} + \frac{1}{3} = 82,$$

since $\gamma = 3^{13}/5^{10} < 1/6$. Hence $R_n \leq 82n^3$ for every $n \geq 1$, which is (7)'s hypothesis. Finally, Shannon entropy dominates order-two Rényi entropy, so $H(C_n) \geq -\log \sum_c p_n(c)^2 = n \log 3 - \log R_n$, whence $D_n \leq \log R_n$ and (7) follows. Dividing by $n \log 3$ gives full entropy dimension. $\square$

6. RIGIDITY OF THE METHOD

It is natural to ask whether Theorem 1.1 is one member of a family. It is not. We show that the hypotheses the proof needs single out the triple $(3, 2, 1/2)$, and that all of them hold with equality there.

Fix coprime integers $p, q \geq 2$ and $\lambda \in (0, 1)$, let the gaps be geometric,

$$w_k = \Pr(K = k) = (1 - \lambda)\lambda^{k-1}, \quad k \geq 1, \quad (24)$$

and consider $X = \sum_{j \geq 0} p^j q^{-s_j} \in \mathbb{Z}_p$, with finite addresses $x_n(w) = \sum_{j < n} p^j q^{-s_j} = N_n(w)/q^n$ as in (9). Lemma 2.1 goes through verbatim with v_2 replaced by v_q , and since $N_n \equiv p^{n-1} \pmod{q}$ for every word of length n , Lemma 2.2 becomes $|x_n(u) - x_n(v)| \geq q p^n / q^n$. The word

probability is $\pi(t) = (1 - \lambda)^n \lambda^{t-n}$, again a function of the total alone. Three quantities then control the argument.

Definition 6.1. With the notation above set

$$\gamma_d = p \sum_{k \geq 1} w_k^2 = \frac{p(1 - \lambda)}{1 + \lambda}, \quad \gamma_m = p \mathbb{E}[q^{-K}] = \frac{p(1 - \lambda)}{q - \lambda}, \quad \gamma_s = \frac{(1 - \lambda)^2 q}{1 - \lambda^2 q},$$

the last defined when $\lambda^2 q < 1$. The hypotheses are

$$(H1) \gamma_d \leq 1, \quad (H2) \gamma_m \leq 1, \quad (H3) \gamma_s \leq 1. \quad (25)$$

Their roles are as follows. (H1) is Lemma 4.1: the diagonal charge equals $(p \sum_k w_k^2)^n$, so it is bounded exactly when the collision entropy of the gap law is at least $\log p$. (H2) is Lemma 4.2: $\mathbb{E}[x_n] = m \sum_{j < n} (pm)^j$ with $m = \mathbb{E}[q^{-K}]$, which is linear in n when $pm = 1$, bounded when $pm < 1$, and exponential otherwise. (H3) controls the shell prefactor. In (15) the second term carries the factor $q^{t-1} \pi(t)$, which is constant in t only if $\lambda q = 1$; in general, summing $2^r \leq x_n$ against the law of T_n yields

$$\sum_{t,r} 2^r q^{t-1} \pi(t) \varepsilon_{t,r} \leq \frac{1}{q} \cdot \frac{1 - \lambda}{1 + \lambda} \sum_{j < n} \gamma_s^{n-1-j} \gamma_d^j, \quad (26)$$

which is polynomial in n precisely when $\gamma_s \leq 1$ and $\gamma_d \leq 1$.

Theorem 6.2 (Rigidity). *Let $p, q \geq 2$ be coprime integers and $\lambda \in (0, 1)$ with $\lambda^2 q < 1$. Then (H1), (H2) and (H3) hold simultaneously if and only if*

$$(p, q, \lambda) = (3, 2, \tfrac{1}{2}),$$

and in that case $\gamma_d = \gamma_m = \gamma_s = 1$.

Proof. Assume (H3). Since $\lambda^2 q < 1$, clearing the denominator gives $(1 - \lambda)^2 q \leq 1 - \lambda^2 q$, that is

$$q[(1 - \lambda)^2 + \lambda^2] \leq 1. \quad (27)$$

The function $\lambda \mapsto (1 - \lambda)^2 + \lambda^2$ is strictly convex on $(0, 1)$ with minimum value $1/2$, attained only at $\lambda = 1/2$. Hence (27) forces $q/2 \leq 1$, so $q = 2$; and then equality $2 \cdot \frac{1}{2} = 1$ is required, so $\lambda = 1/2$ and $\gamma_s = 1$. Note $\lambda^2 q = 1/2 < 1$, consistent.

With $q = 2$ and $\lambda = 1/2$ we get $\sum_k w_k^2 = (1 - \lambda)/(1 + \lambda) = 1/3$, so (H1) reads $p/3 \leq 1$, i.e. $p \leq 3$. Since $p \geq 2$ and $\gcd(p, 2) = 1$ we must have $p = 3$, and then $\gamma_d = 1$.

Finally $\mathbb{E}[q^{-K}] = (1 - \lambda)/(q - \lambda) = \frac{1/2}{3/2} = 1/3$, so $\gamma_m = 3 \cdot \frac{1}{3} = 1$ and (H2) holds, again with equality. Conversely $(3, 2, 1/2)$ satisfies all three, as just computed. $\square$

Remark 6.3. The obstruction is (H3), and it is geometric rather than arithmetic: separation is measured in a single archimedean absolute value, and the resulting prefactor $q^{t-1} \pi(t)$ can be tamed only when the geometric decay of the word probability exactly cancels the growth of q^t . With $\lambda = 1/q$, so that the gap law is $w_k = (q - 1)q^{-k}$, the prefactor equals $(q - 1)^n / q$, bounded only for $q = 2$. A route past this barrier would presumably replace the single real place by a product formula over several places; we do not pursue it.

Remark 6.4. Theorem 6.2 explains why the measure (2) is the one that can be treated: it is the unique point at which the collision entropy of the gap law, the first moment of the real address, and the shell prefactor are all exactly critical. An exhaustive rational search over coprime $p, q \leq 12$ and $\lambda = a/b$ with $b \leq 24$ returns this triple and no other; see Appendix A.

7. THE TRUE ORDER OF R_n

Theorem 1.1 is correct but not sharp. This section records what is true, isolates the two independent losses, and states what would remove them. Nothing here is used elsewhere in the paper.

Because 2 is a primitive root modulo 3^n , the recursion $C_n \equiv 2^{-K}(1 + 3C_{n-1}) \pmod{3^n}$ is a cyclic convolution of length $2 \cdot 3^{n-1}$ in discrete-logarithm coordinates, so p_n and hence R_n are computable to machine precision from that exact recurrence by fast Fourier transform. (The finite-word computations reported later in this section are, by contrast, exact integer arithmetic.) The result is

n	R_n	R_n/n^3	$\mathcal{G}_n$	$82n^3$
4	3.06865	0.04795	0.23276	5248
6	4.00033	0.01852	0.23275	17712
8	4.93174	0.00963	0.23333	41984
10	5.86579	0.00587	0.23408	82000
12	6.80290	0.00394	0.23481	141696
14	7.74277	0.00282	—	225008

where $\mathcal{G}_n = (R_{n+1} - R_n)/2$. A least-squares fit over $7 \leq n \leq 13$ gives $R_n \approx 0.468n + 1.19$ to three decimals, and the increments $\mathcal{G}_n$ drift upward by about one percent over seven levels. The evidence is therefore for

Conjecture 7.1. $R_n = \Theta(n)$, and consequently $D_n \leq \log n + O(1)$.

The two powers of n separating (6) from Conjecture 7.1 are independent and can be located exactly. Write $a_{t,r} = 3^{n/2} \|p_{t,r}\|_2$ as in Proposition 4.4 and define the *effective cell count*

$$\mathcal{Q}_n^{\text{eff}} = \frac{R_n^{\text{core}}}{\sum_{t,r} a_{t,r}^2},$$

so that $R_n^{\text{core}} = \mathcal{Q}_n^{\text{eff}} \sum a_{t,r}^2$ tautologically, and $\mathcal{Q}_n^{\text{eff}} \leq (\sum a_{t,r})^2 / \sum a_{t,r}^2 \leq Q_n$ by Minkowski and Cauchy–Schwarz. Proposition 4.4 replaces $\mathcal{Q}_n^{\text{eff}}$ by the crude Q_n . Exact rational computation over all words with $n \leq t < 6n$ gives

n	Q_n	$35n^2$	$\sum a_{t,r}^2$	$1 + n/6$	$\mathcal{Q}_n^{\text{eff}}$
1	5	35	0.99902	1.1667	1.5865
2	46	140	0.99999	1.3333	2.1251
3	106	315	1.00012	1.5000	2.6004
4	191	560	1.00426	1.6667	3.0548
5	325	875	1.00965	1.8333	3.5002
6	437	1260	1.01892	2.0000	3.9260
7	596	1715	1.02995	2.1667	4.3359

Two things are visible. First, $\sum a_{t,r}^2$ is flat, near 1, whereas Proposition 4.4 uses the bound $1 + n/6$; the shell charge of Lemma 4.2 is never approached, because (15) replaces $\sum_c (\text{count}_c)^2$ by $\max_c (\text{count}_c) \cdot \sum_c \text{count}_c$ and almost every class contains a single word. Second, $\mathcal{Q}_n^{\text{eff}}$ grows linearly, at roughly $0.44n + 1.2$, whereas Cauchy–Schwarz charges $Q_n = \Theta(n^2)$; the mass concentrates on totals within $O(\sqrt{n})$ of $2n$ and on shells within $O(1)$ of $r = -K_0$, so the overwhelming majority of the $35n^2$ cells are negligible.

Each observation, if proved, removes one factor of n :

Conjecture 7.2. $\sum_{t,r} 3^n \|p_{t,r}\|_2^2 = O(1)$, and $\mathcal{Q}_n^{\text{eff}} = O(n)$.

Together they give Conjecture 7.1. We record the obstacles we met. Restricting the ranges of t and r to the $O(\sqrt{n})$ and $O(1)$ windows that carry the mass leaves a complement of only polynomially small probability, and the crude estimate $R^{\text{tail}} \leq 3^n \|p\|_1^2$ used in (23) requires exponentially small mass; both ranges must therefore stay of length $\Theta(n)$, which forces $Q_n = \Theta(n^2)$. Weighted Cauchy–Schwarz does not help either, and it is worth recording exactly why, because the reason locates the obstruction. Split (15) as $a_{t,r} \leq b_{t,r} + c_{t,r}$ with $b_{t,r} = (3^n 2^{-t} \varepsilon_{t,r})^{1/2}$ and $c_{t,r} = (2^{r-1} \varepsilon_{t,r})^{1/2}$. For the diagonal part one can indeed do better than Proposition 4.4: the summand

$$f_t = 3^n 2^{-t} \Pr(T_n = t) = \binom{t-1}{n-1} \left(\frac{3}{4}\right)^n \left(\frac{1}{4}\right)^{t-n}$$

appearing in Lemma 4.1 is itself a negative binomial probability mass function with parameters $(n, 3/4)$, of total mass 1, mean $4n/3$ and standard deviation $\frac{2}{3}\sqrt{n}$. Concentration therefore gives $\sum_t \sqrt{f_t} = O(n^{1/4})$, numerically $1.828 n^{1/4}$, and Cauchy–Schwarz over the at most $7n$ shells of a given total yields

$$\sum_{t,r} b_{t,r} \leq \sqrt{7n} \sum_t \sqrt{f_t} = O(n^{3/4}), \quad (28)$$

against the $O(n)$ that (18) alone provides. But the diagonal part is not the leading term. The same manoeuvre fails for the shell part, because $h_t = \sum_r 2^{r-1} \varepsilon_{t,r} \approx \frac{1}{2} \mathbb{E}[x_n; T_n = t]$ does *not* concentrate: under the x_n -weighted measure the j -th term of (2) contributes near $T_n \approx 2n - \frac{2}{3}j$, so the weight is spread over the whole window $[4n/3, 2n]$, of width $\Theta(n)$. One is left with $\sum_{t,r} c_{t,r} \leq \sqrt{7n} (5n \cdot n/6)^{1/2} = \sqrt{35/6} n^{3/2}$, whose square is $\frac{35}{6} n^3$: exactly the leading constant of Proposition 4.4. The refinement thus gains nothing at all, not even in the constant.

The shell sector splits further, and the split isolates the obstruction completely. For $r < 0$ a geometric weight is available: since $\varepsilon_{t,r} \leq \Pr(x_n < 2^{r+1}) \leq \Pr(K_0 > -r-1) = 2^{r+1}$ and also $\varepsilon_{t,r} \leq \Pr(T_n = t)$,

$$\sum_{r < 0} c_{t,r} \leq \sqrt{\Pr(T_n = t)} \sum_{r < 0} 2^{(r-1)/2} = \frac{1}{2 - \sqrt{2}} \sqrt{\Pr(T_n = t)}, \quad (29)$$

so $\sum_t \sum_{r < 0} c_{t,r} \leq (2 - \sqrt{2})^{-1} \sqrt{5n} = O(\sqrt{n})$; in the exact finite-word enumeration this sum is in fact bounded, converging to about 1.6. The whole difficulty lies in $0 \leq r \leq n \log_2(3/2)$.

There the natural remedy is again a geometric weight, and it is unavailable for a structural reason. Cauchy–Schwarz with $u_{t,r} = 2^{-\theta r}$, $\theta > 0$, gives

$$\sum_{t,r \geq 0} c_{t,r} \leq \left(\frac{1}{2} \mathbb{E}[x_n^{1+\theta}] \right)^{1/2} \left(\frac{5n}{1-2^{-\theta}} \right)^{1/2},$$

so it requires a moment of order $1 + \theta$. No such moment exists.

Lemma 7.3. *The increasing limit $x = \lim_n x_n = \sum_{j \geq 0} 3^j 2^{-S_j}$ is finite almost surely. Moreover $\mathbb{E}[x^p] < \infty$ for every $0 < p < 1$, and $\mathbb{E}[x^p] = \infty$ for every $p \geq 1$.*

Proof. Almost sure finiteness. Since $\mathbb{E}K = \sum_{k \geq 1} k 2^{-k} = 2$, the strong law of large numbers gives $S_j/(j+1) \rightarrow 2$ almost surely. Fix ε with $0 < \varepsilon < 2 - \log_2 3$, which is possible because $\log_2 3 < 2$. Almost surely there is a j_0 with $S_j \geq (2 - \varepsilon)(j+1)$ for all $j \geq j_0$, and then

$$3^j 2^{-S_j} \leq 3^j 2^{-(2-\varepsilon)(j+1)} = 2^{-(2-\varepsilon)} \left(\frac{3}{2^{2-\varepsilon}} \right)^j.$$

The ratio $3 \cdot 2^{-(2-\varepsilon)}$ is less than 1 precisely because $\varepsilon < 2 - \log_2 3$, so the series $\sum_j 3^j 2^{-S_j}$ is dominated by a convergent geometric series beyond j_0 and therefore converges almost surely. (Termwise decay alone would not suffice.)

Finiteness of the moments below 1. Let $0 < p < 1$. Subadditivity of $t \mapsto t^p$ on $[0, \infty)$ gives $x_n^p \leq \sum_{j < n} (3^j 2^{-S_j})^p$, so by independence

$$\mathbb{E}[x_n^p] \leq \sum_{j < n} 3^{jp} \mathbb{E}[2^{-pK}]^{j+1} = \mathbb{E}[2^{-pK}] \sum_{j < n} \left(3^p \mathbb{E}[2^{-pK}]\right)^j.$$

Since $\mathbb{E}[2^{-pK}] = \sum_{k \geq 1} 2^{-(1+p)k} = (2^{1+p} - 1)^{-1}$, the ratio is $3^p \mathbb{E}[2^{-pK}] < 1$ exactly when

$$3^p + 1 < 2^{p+1}, \quad (30)$$

which holds precisely for $p < 1$, with equality at $p = 1$. The geometric series is then bounded uniformly in n , and monotone convergence gives $\mathbb{E}[x^p] < \infty$.

Divergence of the moments from 1 up. For $p \geq 1$ a finite p -th moment would force $\mathbb{E}[x] \leq (\mathbb{E}[x^p])^{1/p} < \infty$ by Lyapunov. But $x \stackrel{d}{=} 2^{-K}(1 + 3x)$ with K independent of x , so a finite mean would satisfy

$$\mathbb{E}x = \mathbb{E}[2^{-K}] (1 + 3 \mathbb{E}x) = \frac{1}{3} + \mathbb{E}x,$$

which is absurd. Hence $\mathbb{E}x = \infty$ and no moment of order $p \geq 1$ is finite. $\square$

Thus x has *critical moment exponent* exactly 1. We do not assert a precise tail asymptotic; only the critical moment statement is used below. The very criticality that produces $\mathbb{E}[x_n] = n/3$ in Lemma 4.2 forbids every geometric shell weight. This is the same phenomenon as the triple equality of Theorem 6.2, appearing a fourth time.

Numerically $\sum b_{t,r}$, $\sum c_{t,r}$ and $\sum a_{t,r}$ grow like $n^{0.45}$, $n^{1.12}$ and $n^{0.84}$ over $3 \leq n \leq 7$, so a bound of order n^2 is presumably true; it is simply not reachable by weighting this decomposition. It is Lemma 4.2 itself, and not the aggregation step, that must be improved. Improving (15) by capping the per-class count at the total number of words in the cell never binds, since the required inequality $n3^n < 2^{2r-1}$ fails for $r \leq 0.585n$. A sharper route appears to need a bound on $\sum_c (\text{count}_c)^2$ directly, which is the quantity being estimated.

Remark 7.4. Identity (31) together with $\mathcal{G}_n \geq 0$ shows that R_n is nondecreasing; the computations above suggest, but do not prove, that $R_n \rightarrow \infty$. If $R_n \rightarrow \infty$, then two consequences follow. First, $D_n \leq \log R_n$ cannot yield $\sup_n D_n < \infty$, so the L^2 method is capped at $D_n = O(\log n)$, which Theorem 1.1 already attains up to the constant. Second, writing Z_n for the density of the level- n discretisation against Haar measure, $R_n = \|Z_n\|_{L^2}^2$, so (Z_n) is unbounded in L^2 and ν has no L^2 density. Absolute continuity with an $L \log L$ density would not be excluded, but neither conclusion is reachable by sharpening the present argument.

8. AN EQUIVALENT AFFINE-CORRELATION FORM

For completeness we record a reformulation. Let $F(x) = 4x + 1$, so $F^h(x) = 4^h x + (4^h - 1)/3$, and put $\Gamma_n(h) = \sum_{y \bmod 3^n} p_n(y) p_n(F^h(y))$.

Lemma 8.1. *With $\text{Col}_n = \sum_c p_n(c)^2$ and $\mathcal{G}_n = 3^n \sum_{h \geq 1} 4^{-h} \Gamma_n(h)$,*

$$\text{Col}_{n+1} = \frac{1}{3} \text{Col}_n + \frac{2}{3} \sum_{h \geq 1} 4^{-h} \Gamma_n(h), \quad \text{equivalently} \quad R_{n+1} - R_n = 2\mathcal{G}_n. \quad (31)$$

Proof. Write $X = 2^{-K}(1 + 3Y)$ and $X' = 2^{-K'}(1 + 3Y')$ with Y, Y' independent copies of X , and let $d = K - K'$, whose law is $\Pr(d) = 1/(3 \cdot 2^{|d|})$.

First, a parity constraint. Modulo 3 we have $2 \equiv -1$, so $X \equiv (-1)^K$ and $X' \equiv (-1)^{K'}$. A collision $X \equiv X' \pmod{3^{n+1}}$ therefore forces $K \equiv K' \pmod{2}$, that is, d even. Odd d contributes nothing.

Second, the reduction. If $d = 0$ the condition becomes $Y \equiv Y' \pmod{3^n}$, of probability Col_n , and $\Pr(d = 0) = \sum_k 4^{-k} = 1/3$. If $d = 2h$ with $h \geq 1$, then $X \equiv X' \pmod{3^{n+1}}$ reads $2^{-2h}(1 + 3Y) \equiv 1 + 3Y'$, i.e. $1 + 3Y \equiv 4^h(1 + 3Y')$, i.e.

$$Y \equiv \frac{4^h - 1}{3} + 4^h Y' = F^h(Y') \pmod{3^n},$$

of probability $\Gamma_n(h)$. Each of the two signs $d = \pm 2h$ has probability $1/(3 \cdot 2^{2h}) = \frac{1}{3}4^{-h}$, so together they contribute $\frac{2}{3}4^{-h}\Gamma_n(h)$. Summing over $h \geq 1$ and adding the $d = 0$ term gives (31). $\square$

Corollary 8.2. *For every $n \geq 1$, $\mathcal{G}_n \leq 41(n+1)^3$, and hence for any cutoff $J(n) \geq 1$, $3^n \sum_{1 \leq h \leq J(n)} 4^{-h}\Gamma_n(h) \leq 41(n+1)^3$.*

Proof. $\mathcal{G}_n \leq R_{n+1}/2 \leq 82(n+1)^3/2$ by (31), $R_n \geq 0$ and Theorem 1.1. The second claim follows by positivity. $\square$

The numerics of Section 7 indicate $\mathcal{G}_n = O(1)$, with $\mathcal{G}_n \approx 0.235$; Corollary 8.2 is far from sharp, and is stated only because the recurrence (31) makes $\mathcal{G}_n$ the natural quantity.

9. WHAT IS AND IS NOT PROVED

The argument uses no fitted state, no numerical extrapolation, and no lower-tail cutoff; every total $n \leq T_n < 6n$, including the typical window $T_n \sim 2n$, is handled by the shell decomposition, and only the exponentially rare tail $T_n \geq 6n$ is removed.

Proved: the polynomial collision bound $R_n = O(n^3)$; $D_n = O(\log n)$; full entropy dimension of ν ; the affine-correlation bound of Corollary 8.2; and the rigidity of the method, Theorem 6.2.

Not proved: $\sup_n D_n < \infty$; absolute continuity of ν with respect to Haar measure; an $L \log L$ density; or $R_n = \Theta(n)$. Full entropy dimension permits a slowly diverging deficit and is strictly weaker than any of these.

Two further limitations of the method should be stated explicitly. Both are *conditional* on $R_n \rightarrow \infty$, which the computations of Section 7 indicate but which this paper does not prove. What (31) does prove, since $\mathcal{G}_n \geq 0$, is that R_n is nondecreasing.

First, the method is capped at $D_n = O(\log n)$. The only route from R_n to the entropy deficit is the Rényi bound $D_n \leq \log R_n$ of Section 5. If $R_n \rightarrow \infty$ then $\log R_n \rightarrow \infty$, and no sharpening of Theorem 1.1 can yield $\sup_n D_n < \infty$. Even the optimal form of the present argument, $R_n = \Theta(n)$ as in Conjecture 7.1, would give only $D_n \leq \log n + O(1)$, which Theorem 1.1 already attains up to the constant and the factor 3. Bounded deficit, and hence absolute continuity by the standard implication, requires a genuinely different mechanism. Note that this limitation concerns the *method* and is insensitive to how large the truth of R_n actually is, provided only that it diverges.

Second, conditionally, ν has no L^2 density. Writing Z_n for the density of the level- n discretisation of ν against Haar measure, $R_n = \|Z_n\|_{L^2}^2$. If $R_n \rightarrow \infty$ then the martingale (Z_n) is unbounded in L^2 and ν admits no square-integrable density. This would not exclude absolute continuity with an $L \log L$ density, a strictly weaker conclusion. We emphasise that the absence of an L^2

density is *not* among the results proved here: it rests on the numerical evidence for divergence recorded in Section 7.

APPENDIX A. ANCILLARY FILES

The directory `anc/` accompanying this article contains a single self-contained verifier, `verify_renewal_entropy.py`, requiring only Python 3.8 and NumPy. It

- computes p_n to machine precision, from the exact recurrence, by the discrete-logarithm convolution of Section 7 and reproduces the table of R_n for $n \leq 14$, together with D_n and $\mathcal{G}_n$;
- verifies the recurrence (31) to machine precision for $n \leq 9$;
- enumerates all renewal words with $n \leq t < 6n$ for $n \leq 7$ in exact integer arithmetic and checks Lemma 3.1 atom by atom, Lemma 4.1, Lemma 4.2, (18) and Proposition 4.4, reproducing the second table of Section 7;
- performs the exhaustive rational search behind Theorem 6.2 over coprime $p, q \leq 12$ and $\lambda = a/b$ with $b \leq 24$.

No statement in Sections 2–6 depends on these computations; they reproduce the constants and the tables.

REFERENCES

- [1] W. Abram and J. C. Lagarias, *p-adic path set fractals and arithmetic*, J. Fractal Geom. **1** (2014), 45–81.
- [2] E. Breuillard and P. P. Varjú, *On the dimension of Bernoulli convolutions*, Ann. Probab. **47** (2019), 2582–2617.
- [3] E. Breuillard and P. P. Varjú, *Entropy of Bernoulli convolutions and uniform exponential growth for linear groups*, J. Anal. Math. **140** (2020), 443–481.
- [4] K. G. Hare and T. Vávra, *Self-similar sets and measures in the p-adics*, J. Fractal Geom. **11** (2024), no. 3/4, 247–287.
- [5] M. Hochman, *On self-similar sets with overlaps and inverse theorems for entropy*, Ann. of Math. (2) **180** (2014), 773–822.
- [6] R. D. Mauldin and M. Urbański, *Dimensions and measures in infinite iterated function systems*, Proc. London Math. Soc. (3) **73** (1996), 105–154.
- [7] E. Mihailescu and M. Urbański, *Hausdorff dimension of the limit set of conformal iterated function systems with overlaps*, Contemp. Math. **600** (2013), 273–290.
- [8] E. Mihailescu and M. Urbański, *Random countable iterated function systems with overlaps and applications*, Adv. Math. **298** (2016), 726–758.
- [9] A. Rapaport, *Proof of the exact overlaps conjecture for systems with algebraic contractions*, preprint, arXiv:2001.01332.
- [10] A. Salehi Golsefidy, *Sum-product phenomena: p-adic case*, J. Anal. Math. **142** (2020), 349–419.
- [11] P. Shmerkin, *On Furstenberg’s intersection conjecture, self-similar measures, and the L^q norms of convolutions*, Ann. of Math. (2) **189** (2019), 319–391.
- [12] P. P. Varjú, *On the dimension of Bernoulli convolutions for all transcendental parameters*, Ann. of Math. (2) **189** (2019), 1001–1011.
- [13] P. P. Varjú, *Absolute continuity of Bernoulli convolutions for algebraic parameters*, J. Amer. Math. Soc. **32** (2019), 351–397.
- [14] P. P. Varjú, *Entropy rates in the dimension theory of self-similar measures*, preprint, arXiv:2509.22042.

INDEPENDENT RESEARCHER

Email address: eric@ericgrover.com